

Jury Q&A preparation

Opening · general questions · tricky questions · key numbers to know by heart

1. How to open (first 45 seconds)

Goal: the jury must know in one breath what RAQIB does, for whom, and the one number that proves it. Say it slowly, look at the jury, not at the screen. Learn it by heart.

“Good morning. We are team RAQIB — *raqib* means *overseer* in Arabic — and we worked on challenge T2, automated targeting of customs controls.

Every day, customs can physically inspect only a small share of declarations. The question is simple: **which ones?** Today, even the best rule we could test finds fraud in only 55% of the inspections it chooses, and it misses most dangerous goods.

RAQIB is an AI assistant for the risk officer. Every morning it ranks all declarations by duty-fraud risk and public-safety risk, fills the day's inspection capacity, and explains each choice in plain language. **It never decides: the officer does.**

On three months of data it had never seen, with **exactly the same 470 inspections**, RAQIB catches **317 frauds instead of 259**, and **41 public-safety threats instead of 7**. Let us show you how.”

French version (if the jury prefers French):

« Bonjour. Nous sommes l'équipe RAQIB — *raqib* veut dire *celui qui veille* — et nous avons travaillé sur le défi T2, le ciblage automatisé des contrôles douaniers.

Chaque jour, la douane ne peut contrôler physiquement qu'une petite partie des déclarations. La question est simple : **lesquelles ?** Aujourd'hui, la meilleure règle que nous avons testée ne trouve une fraude que dans 55 % des contrôles qu'elle choisit, et laisse passer la plupart des marchandises dangereuses.

RAQIB est un assistant IA pour l'agent de ciblage. Chaque matin, il classe toutes les déclarations selon le risque de fraude et le risque pour la sécurité publique, remplit la capacité de contrôle du jour et explique chaque choix en langage clair. **Il ne décide jamais : c'est l'agent qui décide.**

Sur trois mois de données jamais vues, avec **exactement les mêmes 470 contrôles**, RAQIB détecte **317 fraudes au lieu de 259** et **41 menaces pour la sécurité au lieu de 7**. Laissez-nous vous montrer. »

Opening do's and don'ts

- **Do** start with the problem and the result, not with the technology. No “LightGBM”, “AUC” or “out-of-fold” in the first minute.
- **Do** say “advisory — the officer decides” early. Customs juries worry about automation replacing judgement.
- **Do** say “public, synthetic data” yourself, before the jury does. Owning the limit makes you credible.
- **Don't** read the slides. **Don't** apologise. **Don't** say “this is just a prototype” in a weak tone — say “a working prototype”.

How to answer any question

- **Answer first, then one number, then stop.** One sentence of answer, one fact, one sentence of bridge back to the value. Max 30–40 s.
- **If you don't know, say so:** “We did not measure that; here is how we would.” Never invent a figure — every number we show is computed by the code.
- **Agree with the true part of a criticism** (“Yes, the data is synthetic...”), then give the answer (“...which is why we only claim the gain over the rule, and propose a shadow-mode pilot”).

- **One speaker per question.** Engineers take model/data/tech questions; the finance members take impact, duties, cost and deployment. Others stay silent unless called.
- **Offer the demo** when it helps: “Can I show you in 10 seconds?” — the inspector page answers most explainability questions.

2. General questions (expect most of these)

Short, confident answers. Blue = general; orange (next section) = tricky.

Q1. In one sentence, what does RAQIB do?

It tells customs, every day and within a fixed inspection capacity, which declarations to inspect (red), check on documents (yellow) or release (green), with four plain-language reasons for each — and the officer makes the final decision.

Best speaker: whoever opened

Q2. Why AI? Why not simply improve the existing rules?

A rule looks at one criterion at a time (for example the product's history). The model combines the history of the product, importer, declarant, seller, origin and office at once. On the same data, with the same capacity, that combination catches 58 more frauds (+22%) and almost six times more safety threats than the best rule.

Best speaker: engineer

Q3. What data did you use?

Public Customs Import Declaration Datasets from the Korea Customs Service and the Institute for Basic Science, MIT licence: 54,000 declarations synthesised from real inspected ones, 22 attributes each. We learn on 15 months (45,519 declarations) and test on the next 3 months (8,481), which the model never saw.

Watch out: say “synthetic” yourself. • **Best speaker:** engineer (data)

Q4. Why didn't you use Tunisian data?

Tunisian inspection data is confidential and the hackathon rules forbid accessing administration systems. We used public data to prove the method; the same pipeline retrains on Tunisian data in minutes, inside the administration. We do show public Tunisian UN Comtrade statistics in the app, but only as context — never as a model input.

Best speaker: engineer

Q5. How does the model work, in simple words?

It learns from past inspections how often each product, importer, seller, origin and office was caught before, and how that combines with value, weight and tax rate. Two models give two probabilities — duty fraud and public safety — and the riskiest cases fill the day's capacity.

Watch out: no jargon unless asked; the technical jury member will follow up. • **Best speaker:** engineer

Q6. How do you measure success? Why not just accuracy?

Accuracy is misleading: customs can only inspect a few percent. What matters is **how many frauds you find with the inspections you can actually do**. So we measure precision at 5% (share of inspections that find fraud) and, for safety, recall at 5% (share of threats found). We also replay the 91 test days with the same daily capacity for every policy.

Best speaker: engineer

Q7. What is the concrete gain?

Same 470 inspections: 317 frauds vs 259 for the best rule (+22%), 41 safety threats vs 7. Or, seen the other way: RAQIB reaches the rule's result with 19% fewer inspections (379 instead of 470). The AI beats the rule in 12 of the 13 test weeks.

Best speaker: finance member

Q8. What does that mean in money?

The Impact page estimates the duties at stake on the frauds caught (declared value × the dataset's tax rate, plus 19% VAT as an assumption), for AI vs rule vs random, in TND. It is illustrative, because the values are synthetic Korean amounts converted to TND. The robust claim is the relative gain; the real amount must be measured in a pilot.

Watch out: do not quote a TND figure as real revenue. • **Best speaker:** finance member

Q9. How does an officer use it day to day?

They open the worklist: every declaration has a lane, a probability and four reasons. They click one, see the explanation and what the current rule would have done, then press Inspect, Document check or Release. The decision is recorded in the journal.

Watch out: offer the demo here. • **Best speaker:** platform engineer

Q10. How do you explain a decision to an officer or a trader?

Each declaration shows the four strongest reasons with the real numbers — for example “this product was fraudulent in 36% of its 127 past declarations, against 22% on average” — plus an exact breakdown from the base rate to the final probability. The main fraud model is a glass box (EBM), as accurate as the black box: AUC 0.771 vs 0.770.

Best speaker: engineer

Q11. What infrastructure does it need? What does it cost?

A normal laptop CPU. The whole pipeline retrains in about 15 seconds on our data, scoring a declaration takes about 0.1 s, and the demo runs fully offline — no cloud, no GPU. All libraries are open-source with permissive licences, so no licence fees.

Watch out: the optional local LLM needs a small GPU (4 GB); it is not required. · **Best speaker:** platform engineer

Q12. How would it connect to the existing customs IT system?

Through a simple API: the clearance system sends a declaration, RAQIB returns a lane, probabilities and reasons. It is read-only and advisory, so it can run beside the current selectivity rules without changing them. We did not build that integration — it is step one of a pilot.

Watch out: don't claim you integrated with any real system. · **Best speaker:** platform engineer

Q13. What is the role of the language model (LLM)?

Optional and limited. A small local model (Qwen3-4B) rephrases already-computed facts into a short brief in French, English or Arabic, and turns a typed question into a worklist filter. It never scores, ranks or decides; a guard rejects any number that is not in the facts, and without it everything works with templates.

Best speaker: engineer

Q14. What would you need to go further?

Three things: anonymised historical inspection outcomes, one pilot office, and a customs risk analyst as partner. Then shadow mode for 4–8 weeks and a controlled pilot with a control group.

Best speaker: finance member

3. Tricky questions (prepare these most)

These test whether you understand your own limits. Stay calm, agree with what is true, then answer.

Q15. Your data is synthetic and Korean. Why should we believe any of this applies to Tunisia?

You are right that the absolute numbers will not transfer. What transfers is the method and the comparison: on the same data and the same capacity, the model beats the best rule by 17.8 points of precision, with a 95% confidence interval of +12.5 to +22.5. That is exactly why we propose a shadow-mode pilot on Tunisian data before any real use.

Best speaker: Yassine / lead engineer

Q16. A 22% fraud rate is unrealistic. Aren't your precision figures inflated?

Yes. The dataset only contains inspected declarations, so fraud is far more frequent than in reality (21.6%). Absolute precision is optimistic — we say so in our limits. That is why we compare everything to the rule and to random on the same data: the gain is the claim, not the 72%.

Best speaker: engineer

Q17. Your “current rule” is a straw man. Real customs use much richer risk profiles.

It is the best of the rules we could build from the data — product history — and it is much stronger than random or importer history (53% vs 19–21% precision). We had no access to real Tunisian risk profiles. In a pilot, the comparison must be against the real selectivity rules, and that is how we designed it.

Watch out: don't claim the rule is what Tunisian customs use. · **Best speaker:** engineer

Q18. An AUC of 0.77 is not very high.

For fraud detection on declarations it is a useful level, and AUC is not what customs experiences. What matters is the top of the list: among the top 1%, 85% are frauds, and at 5% capacity 72% vs 53% for the rule. For public safety the AUC is 0.95.

Best speaker: engineer

Q19. Only 73 critical cases in the test set — is the safety result statistically meaningful?

It is small, and we say ± 5 points. But the gain is large: recall at 5% is 78% vs 55%, confidence interval +13 to +32 points, still clearly positive. In the replay, 41 threats vs 7. A pilot with more data must confirm it.

Best speaker: engineer

Q20. You learn only from inspected declarations. Isn't that biased?

Yes — selection bias is a real risk: the model only learns where officers already looked. Two answers: an optional 10% random “exploration” share of inspections, and a random control group in production. Exploration costs 23 frauds in the replay but widens coverage from 261 to 280 distinct products.

Best speaker: engineer

Q21. Fraudsters will learn your model and adapt.

Partly true for any system. Three protections: the features are history-based, so a new behaviour shows up quickly once caught; weekly monitoring and retraining; and the random exploration share makes selection unpredictable. And the officer always keeps discretion.

Best speaker: engineer

Q22. Isn't using the country of origin or the importer discriminatory?

Origin and operator history are standard customs risk criteria, and the model uses the observed history, not nationality or identity of people. We monitor selection rates by office and transport mode, the officer decides, and every decision is logged. If a criterion is judged unacceptable, we can remove it and measure the cost — the pipeline makes that a one-line change.

Watch out: stay factual, don't get defensive. · **Best speaker:** finance member or engineer

Q23. If the AI is wrong and a dangerous shipment passes, who is responsible?

The officer and the administration, exactly as today: RAQIB is advisory. It gives a probability and reasons; the officer decides and the decision is recorded in a tamper-evident journal. It also reduces risk: when its two models disagree, it never releases — it sends the case to a human (371 cases, 44% fraud).

Best speaker: finance member

Q24. What about personal data protection?

The model uses business identifiers (importer, declarant, seller codes), not personal data about individuals. In production it is retrained and run inside the administration, offline, so no data leaves. Governance: compliance with Organic Law 2004-63 and review by the INPDP before any pilot.

Watch out: don't claim legal certainty — say “to be validated with the INPDP”. · **Best speaker:** finance member

Q25. You set the lane thresholds on the test period. Isn't that cheating?

Good catch, and we list it in our limits. It affects only where the red/yellow cut-offs sit, not the ranking or the model scores, which never saw the test data. In production the thresholds come from the previous weeks.

Best speaker: engineer

Q26. Why not deep learning, graph neural networks, or anomaly detection?

We tested the ideas that made sense and kept only what worked. Anomaly detection (Isolation Forest) had no signal (AUC 0.48). Network features gave no gain (0.7690 vs 0.7696). On tabular declarations, gradient boosting and EBM are the state of the art — and EBM is transparent.

Best speaker: engineer

Q27. New importers have no history. What do you do with them?

About 10% of test declarations come from never-seen importers. The model falls back on the other keys — product, HS family, seller, origin, office — and treats the unknown importer as average. In production we could add registry information such as company age.

Best speaker: engineer

Q28. Your value and weight features may be an artefact of the synthetic data.

Possibly: 78% of test declarations have exactly their product's usual unit value, which is unrealistic. Without value and mass, precision drops from 71.1% to 63.3% — still above the rule's 53.4%. We flag it as something to re-validate on real data.

Best speaker: engineer

Q29. Did you use AI to write your code?

Yes, we used Claude Code as a coding assistant, and we say so in our documents. We designed the approach, chose the data and the evaluation, and checked the results. Every number in the app and the slides is recomputed by the code from the data — nothing typed by hand.

Watch out: answer plainly; hiding it is worse. · **Best speaker:** Yassine

Q30. Exploration wastes inspections. Why accept losing 23 frauds?

It is an insurance premium. Without it the model only ever sees what it already suspects and slowly goes blind. It also gives an unbiased measurement of performance. It is a setting: customs can choose 5% or 10%.

Best speaker: finance member

Q31. How is this different from existing tools like ASYCUDA's selectivity or the WCO's DATE model?

Our dataset comes from the WCO BACUDA ecosystem that produced DATE, so we build on that line of work. What we add: allocation under a fixed daily capacity, two objectives (revenue and safety), a glass-box model with plain-language reasons, the current rule shown side by side, uncertainty sent to a human, and a tamper-evident journal.

Watch out: don't claim what other systems cannot do. · **Best speaker:** engineer

Q32. Why should customs trust a student prototype?

They shouldn't trust it blindly — that is why we propose shadow mode first: RAQIB runs in parallel for 4–8 weeks, changes no decision, and its choices are compared to the real outcomes. Trust comes from measurement.

Best speaker: Yassine

Q33. The LLM could hallucinate a wrong reason to an officer.

That is why it only rephrases facts the models already computed, a number guard rejects any figure not in the facts, accusatory wording is banned, and otherwise the deterministic template is used. It never scores or decides.

Best speaker: engineer

Q34. What happens if RAQIB is down?

Nothing breaks: it is advisory, so the current rules keep working. It runs offline on a single machine, with no cloud dependency.

Best speaker: platform engineer

4. Numbers to know by heart

If you quote a number, it must be one of these. All from the test period (Apr–Jun 2021), never used for training.

Topic	Number	Say it like this
Replay, same capacity	470 inspections · 91 days · 5%/day	"Same 470 inspections for every policy."
Frauds caught	317 AI · 259 rule · 103 random	"58 more frauds, +22%."
Safety threats caught	41 AI · 7 rule · 3 random (of 73)	"Almost six times more threats."
Efficiency	379 vs 470 inspections for the rule's result	"19% fewer inspections."
Precision @5%	71.8% vs 53.4% (CI +12.5 to +22.5 pts)	"+18 points, significant."
Safety recall @5%	78.1% vs 54.8% (CI +13 to +32 pts)	"+22 points, small sample."
Top 1%	84.7% are frauds	"In the top 1%, 85 of 100 are frauds."
Stability	12 of 13 weeks	"Not a lucky week."
Calibration	61.7% predicted vs 61.4% observed (top tenth)	"The probabilities are honest."
Glass box	EBM 0.771 vs LightGBM 0.770 AUC	"Transparent, no accuracy lost."
Disagreement → human	371 cases (4.4%), 44% fraud vs 22% avg	"When unsure, it asks a human."
Green lane	82% released	"Faster clearance for honest traders."
Data	54,000 decl. · 22 attributes · 45,519 train / 8,481 test	"Public, synthetic, MIT licence."
Speed	~15 s full retrain · ~0.1 s per declaration	"Runs on a laptop, offline."

Closing line (if you get the last word)

"With the same officers and the same inspections, RAQIB finds more fraud and far more dangerous goods — and the officer stays in charge. We are asking for one thing: a shadow-mode pilot in one office, measured against today's rules. Thank you."